DATA SCIENCE- Methodology

Analysis of patterns of dead projects on GitHub

This project follows a structured end-to-end data science workflow to analyze the lifecycle and decline patterns of open-source repositories hosted on GitHub. The objective is to identify measurable indicators that signal decreasing project health and long-term inactivity.

Step 1: Problem Definition

The research problem focuses on identifying behavioural and structural patterns that precede project inactivity. Project decline is defined through measurable metrics such as reduced commit frequency, declining contributor participation, and increasing issue resolution time.

Step 2: Data Collection

Repository metadata will be collected using the GitHub REST API or publicly available opensource datasets. Repositories from diverse domains and activity levels will be selected to ensure variability. Data fields include commit timestamps, contributor count, issue and pull request statistics, release frequency, repository age, and activity logs.

Step 3: Data Preprocessing

The collected dataset will be cleaned to remove duplicates and incomplete records. Missing values will be handled appropriately. Time-based data will be standardized, and new features such as commit velocity (commits per month), contributor churn rate, and issue resolution ratio will be engineered.

Step 4: Exploratory Data Analysis (EDA)

Statistical summaries and visualizations such as time-series plots, histograms, boxplots, and correlation heatmaps will be generated. This step identifies trends in contributor engagement, maintenance slowdown, and structural instability.

Step 5: Trend & Comparative Analysis

Repositories will be categorized based on activity levels (active, moderately active, declining). Time-based analysis will help detect early warning signals that precede inactivity.

Step 6: Interpretation & Reporting

Findings will be interpreted in the context of project sustainability and organizational dependency risk. Ethical considerations and limitations will be discussed to ensure responsible conclusions.

2. Hardware and Software Requirements

Hardware Used

- Processor: Intel Core i5 (or equivalent)
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or higher
- Internet connection for API access and dataset retrieval

Software Used

- Operating System: Windows 10/11
- Programming Language: Python 3.x
- Development Environment: Jupyter Notebook / VS Code
- Libraries:
    - Pandas (data manipulation)
    - NumPy (numerical computation)
    - Matplotlib & Seaborn (visualization)
    - Requests (API handling)
    - Scikit-learn (if required for modeling)

Flowchart


PROBLEM DEFINITION
Define objectives and identify decline metrics
REPOSITORY SELECTION
Select diverse GitHub repositories based on size, age, and activity
DATA COLLECTION
Extract repository metadata using GitHub API / Public datasets
DATA CLEANING & PREPROCESSING
Remove duplicates, handle missing values, standardize timestamps
FEATURE ENGINEERING
Compute commit velocity, contributor churn, issue resolution rate, release frequency
EXPLORATORY DATA ANALYSIS (EDA)
Time-series plots, heatmaps, distributions
TREND & COMPARATIVE ANALYSIS
Compare active vs declining repositories
INSIGHT GENERATION & INTERPRETATION
Identify early warning indicators
Assess sustainability risk
ETHICS, BIAS & LIMITATIONS
Address survivorship bias & data constraints
CONCLUSION & REPORT